Contour Detection and Image Segmentation

* Study of different contour detection and image segmentation algorithms

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***Abstract*—In This study, we investigated two fundamental problems in computer vision: contour detection and image segmentation. we studied 4 different algorithms, two algorithms for each problem. For contour detection, we implemented PbLite algorithm which is an simplied version of gPb algorithm presented in [1] and We trained an encoder decoder network using PASCAL VOC data-set. For image segmentation, we compared the results of segmentation using Mean-shift clustering algorithm and Pre-trained FCN-ResNet101 model.**

***Index Terms*—contours, image processing, computer vision, contour detection, image segmentation, deep learning, encoder decoder convolutional neural networks**

I. Introduction

Contour detection and image segmentation are two fundamental problems in computer vision. First , Contour detection is one of the most controversial subject nowadays as it is used in a lot of applications like background removal and object detection . For now , there is no a perfect state of art for contour detection . Also , most contour detection techniques output a lot of noise . So we tried two techniques for contour detection which are Pblite that is non-machine learning and Encoder-decoder algorithm that is machine learning.

Second , Segmentation has a lot of techniques to manipulate some are deep learning based and another are not deep learning based . So we compared two different techniques which are FCN-ResNet that is deep learning based and Mean-Shift Clustering that is machine learning based.

In this research , we are going to go deep into contour detection and image segmentation background in section two (II) . Moreover , Implementation would be explained deeply in section three (III) . After that , Evaluation would be held to compare between the two techniques in each topic in section four (IV) . Furthermore , we would recap all the research results in the conclusion in section five (V).

II. Background

Contours

The problems of contour detection and image segmentation are related problems, but not identical. contour detectors do not offer guarantee that they will produce closed contours and hence do not provide a partition of the image into regions. But, one can recover closed contours from regions in the form of the boundaries. Historically, there have been different lines of approach to these two problems.

Early approaches to contour detection aim at quantifying the presence of a boundary at a given image location through local measurements. The Roberts, Sobel, and Prewitt operators detect edges by convolving a gray-scale image with local derivative filters. Marr and Hildreth use zero crossings of the Laplacian of Gaussian operator. The Canny detector [2] also models edges as sharp discontinuities in the brightness channel, adding non-maximum suppression and hysteresis thresholding steps. A richer description can be obtained by considering the response of the image to a family of filters of different scales and orientations. An example is the Oriented Energy approach which uses quadrature pairs of even and odd symmetric filters. Lindeberg proposes a filter-based method with an automatic scale selection mechanism.

There is also more recent approaches that take into account color and texture information and make use of learning techniques for cue combination. For example, Martin et al. [3] define gradient operators for brightness, color, and texture channels, and use them as input to a logistic regression classifier for predicting edge strength.

There are multiple of different approaches introduced in recent papers. In Arbeláez et al. [1], The authors introduced state of the art contour detection algorithm that combine local and global image information. Their contour detector combines multiple local cues into a globalization framework based on spectral clustering. The gPb contour detector performed significantly better than other algorithms scoring F1-measure of 0.7 on the Berkeley Segmentation Dataset (BSDS300).

The Convolutional neural networks has offered significant performance in extracting the features of the image and Because of the evolution in the computational power, more researchers are eager to use the deep learning techniques in solving problems that conventional techniques can not solve. In Yang et al [5], The authors developed a deep learning algorithm for contour detection with a fully convolutional encoder-decoder network. They trained their network end-to-end on PASCAL VOC with refined ground truth from inaccurate polygon annotations, yielding much higher precision in object contour detection than previous methods. their method generated high-quality segmented object proposals, which significantly advance the state-of-the-art on PASCAL VOC (improving average recall from 0.62 to 0.67) with a relatively small amount of candidates (around 1660 per image).

In [9], The authors proposed a convolutional encoder–decoder framework to extract image contours supported by a generative adversarial network to improve the contour quality. the proposed generative adversarial network they used aimed to increase the detection accuracy. The resulting method contained two models, an encoder–decoder model, whose weights were updated using binary cross-entropy loss with fine-tuning from the VGG16 pre-trained model, and a discriminator network that employed ground truths and predicted contours as input for discrimination. They evaluated the method on the Berkeley Segmentation DataSet and Benchmarks 500 and NYUD dataset, achieving state-of-the-art performance with an F1-measure of 0.810 and 0.715.

Segmentation

A broad family of approaches to segmentation involve integrating features such as brightness, color, or texture over local image patches and then clustering those features based on fitting mixture models, mode-finding, or graph partitioning. Three algorithms in this category appear to be the most widely used as sources of image segments in recent applications, due to a combination of reasonable performance and publicly available implementations.

In Felzenszwalb et al [6], The authors presented The graph based region merging algorithm. The algorithm tempts to partition image pixels into components such that the resulting segmentation is neither too coarse nor too fine.

The Mean Shift algorithm [7] offers different clustering framework. The pixels are represented in the joint spatial-range domain by concatenating their spatial coordinates and color values into a single vector. Applying mean shift filtering in this domain give a convergence point for each pixel. Regions are formed by grouping together the pixels whose convergence points are closer than h_s in the spatial domain and h_r in the range domain, where h_s and h_r are respective bandwidth parameters.

In Arbeláez et al. [1], In addition to the gPb contour detector, the authors also introduced new state-of-the-art segmentation algorithm. their segmentation algorithm consists of generic machinery for transforming the output of any contour detector into a hierarchical region tree. In this manner, they reduced the problem of image segmentation to that of contour detection. By feeding the output from the gPb contour detector as input to the segmentation algorithm, they produced regions whose boundaries match ground-truth better than those produced by many other methods. In scale of F1-measure, the algorithm scored 0.71 on the the BSDS300 Benchmark.

As reviewed in [10], the authors investigated using data augmentation approach to balance the semantic label distribution in order to improve segmentation performance. the authors proposed using generative adversarial networks (GANs) to generate realistic images for improving the performance of semantic segmentation networks. Experimental results showed that the proposed method can not only improve segmentation performance on those classes with low accuracy, but can also obtain 1.3% to 2.1% increase in average segmentation accuracy. They concluded that augmentation method can boost accuracy and be easily applicable to any other segmentation models.

in [11], the author reviewed the field of semantic segmentation as pertaining to deep convolutional neural networks. the author provided comprehensive coverage of the top approaches and summarized the strengths, weaknesses and major challenges. The author found that recent work based largely on deep learning techniques which has resulted in groundbreaking improvements in the accuracy of the segmentation (reported accuracy over 79% (mIOU) on the PASCAL VOC-2012 test set).

III. Implementation

For contour detection, we used two different algorithm to detect the contour. First approach was to use non machine learning algorithm. We used PbLite algorithm, which is a simplified version of gPb algorithm implemented in [1]. Second approach, we tried different flow, which is using deep learning techniques. We followed the same network used in [5].

PbLite Contour Detector

The most naive ways for detecting the contours is using simple edge detection methods for example Sobel or Canny. The problem of these algorithms it only take into consideration the intensity values. The PbLite algorithm is a simpler version of gPb algorithm designed by Arbelaez et al [1].

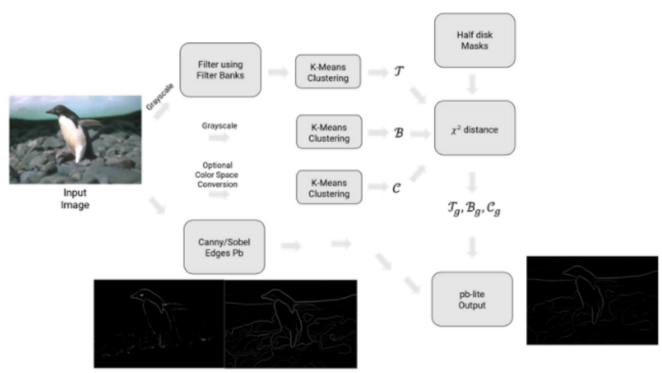

Fig. 1. Pb Lite algorithm

AS shown in figure 1, The PbLite algorithm runs through different steps.

Generate the filter banks: The first step in the algorithm is to create a set of filters with which the image of interest is filtered to produce useful responses. A filter bank is defined as a collection of filters. There are three types of filter banks used for this process:

- Oriented DoG Filters: A collection of oriented Derivative of Gaussian filters is a simple yet powerful collection of filters. These can be generated as the result of the convolution on a Gaussian kernel with a Sobel filter.
- Leung-Malik (LM) Filters: Another set of filters are the multi scale, multi orientation filter bank, which consist

of the Leung- Malik filters. LM filter bank consists of first and second order derivatives of Gaussians at 6 orientations and 3 scales making a total of 36; 8 Laplacian of Gaussian filters and 4 Gaussians.
- Gabor Filters: Filters derived from the ones present in the human visual system, the Gabor filter bank is a Gaussian kernel function modulated by a sinusoidal plane wave.

Image processing: After creating the set of filters. The second phase is to start processing the input image. The image is processed through these steps:

- New version of the image is to be created and converted to gray-scale.
- Generate Texton Map($\mathcal{T}$): To generate the texton map, the input image is filtered with each filter of the filter bank. This gives N filter responses at each pixel of the image. Hence we obtain an N dimensional vector on which we apply a K-Means clustering with the number of clusters K = 64.
- Generate Brightness Map ($\mathcal{B}$): The Brightness Map B is generated by performing a KMeans clustering on the gray-scale equivalent of the color image with the cluster size K = 16.
- Generate Color Map ($\mathcal{C}$): Color Map C captures the changes in the chrominance in the image. It is generated by performing a K-Means clustering on the color image with the cluster size K = 16.
- After Generating the three maps, We next step is to generate their gradients ($\mathcal{T}_g$, $\mathcal{B}_g$, $\mathcal{C}_g$). The three gradients indicate how much the texture, brightness and color distributions are changing at a pixel. These gradients are found out by comparing the distributions in left and right of the half disc pairs which gives us the χ^2 measure, which is simply the sum of squared difference between histogram elements along with a soft normalization so that less frequent elements still contribute to the overall distance.
- The last step of image processing is to apply the Soble and Canny edge detection on the image to find the edges.

Generate the output: The last step of the pipeline is to combine information from the features (Texture, Brightness and Color Gradients $\mathcal{T}_g$, $\mathcal{B}_g$, $\mathcal{C}_g$) with the baseline outputs from the Sobel or Canny edge. The output is generated using the equation 1:

$$PbEdges = \frac{\mathcal{T}_g + \mathcal{B}_g + \mathcal{C}_g}{3} \cdot w1 * canny + w2Sobel) \quad (1)$$

A. Encoder Decoder Network

Given the success of deep convolutional networks for learning rich feature hierarchies, many researchers aimed to use deep learning techniques to solve image problem. The problem of finding the contours can be formulated to be a labelling problem (e.g 0 or 1). Encoder-decoder networks is type of Convolutional neural networks (CNNs) which uses a encoder for example (VGG16) to extract the features of the image, then uses un-polling and de-convolution layer to generate the output image. In this project, we used the VGG16 model as encoder.

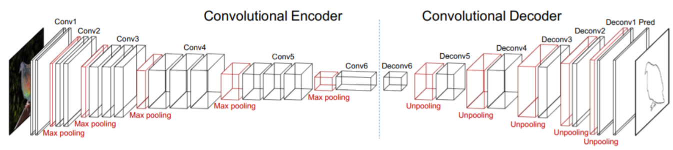

Fig. 2. Encoder Decoder Network Architecture

VGG16 is a convolutional neural network model proposed by K. Simonyan and A. Zisserman from the University of Oxford in the paper "Very Deep Convolutional Networks for Large-Scale Image Recognition" [8]. The model achieves 92.7% top-5 test accuracy in ImageNet, which is a dataset of over 14 million images belonging to 1000 classes. VGG16 model makes the improvement over AlexNet by replacing large kernel-sized filters with multiple 3×3 kernel-sized filters one after another.

For the decoder network, The first layer of decoder "deconv6" is designed for dimension reduction that projects 4096-d "conv6" to 512-d with 1×1 kernel so that the pooling switches from "conv5" can be re-used to upscale the feature maps by twice in the "deconv5" layer. The number of channels of every decoder layer is properly designed to allow unpooling from its corresponding maxpooling layer. All the decoder convolution layers except "deconv6" use 5×5 kernels. All the decoder convolution layers are followed by relu activation function except the one next to the output label. A complete decoder network setup is shown in Table I. the loss function is the pixel-wise logistic loss.

name	setup kernel activation
deconv6	Conv 1x 1 x 512 ReLU
deconv5	unpool-conv 5x5x512 ReLU
deconv4	unpool-conv 5x5x256 ReLU
deconv3	unpool-conv 5x5x128 ReLU
deconv2	unpool-conv 5x5x64 ReLU
deconv1	unpool-conv 5x5x32 ReLU
pred	conv 5x5x1 sigmoid

TABLE I
DECODER ARCHITECTURE

We trained the network using the PASCAL VOC dataset for 30 epochs using Adam optimizer with learning rate 10e-4.

For image segmentation, We tried also two different algorithms. The first approach is to use meanshift clustering [7] and the second apporach is to use a pre-trained version of FCN-ResNet101 encoder decoder network.

Segmentation using Mean Shift Clustering

To perform segmentation using meanshift clustering algorithm, The following steps were performed:

- The image need to be stored into a dataframe in the format of (row, col, r, g, b).
- the values needs to be normalized in order to get best results.

- the bandwidth values is estimated using sklearn library.
- The meanshift algorithm is used to cluster the values and gets the centers of each segment.
- after segmentation is done the image were reshaped to the initial shape.

After the clustring is done and to remove the noise, we applied a median filter of kernel size 3x3 to remove the noise.

Segmentation using FCN-ResNet101

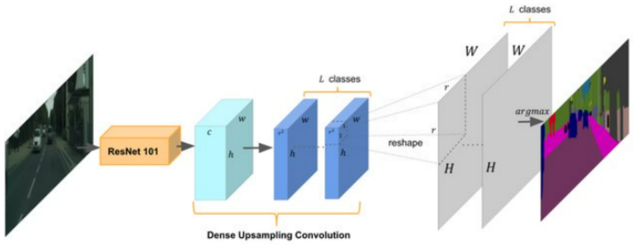

Fig. 3. FCN-ResNet101 architecture

FCN-ResNet is constructed by a Fully-Convolutional Network model, using a ResNet-50 or a ResNet-101 backbone. The pre-trained models have been trained on a subset of COCO train2017, on the 20 categories that are present in the Pascal VOC dataset. We choose ResNet101 because it has 101 layer which are better to detect more features of the image. The architecture of FCN-ResNet101 is shown in figure 3. The FCN-ResNet101 scored average accuracy of 30.84 on the PASCAL VOC Evaluation Server2012.

IV. Evaluation

Contour Detection

After Training the encoder decoder network, we used the network to predict the contours for 200 test images in the BSDS500 dataset. The same for the PbLite algorithm. We used PSNR as evaluation metric. The two algorithms scored almost the same average PSNR.

Although both algorithms scored the same scores, there is huge difference between the output of each algorithm.

Fig. 4. First Example

Fig. 5. Second Example

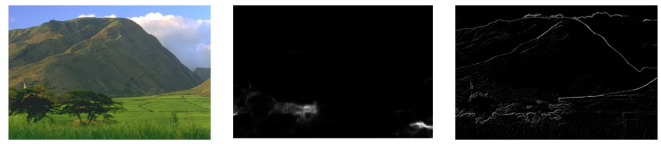

Fig. 6. Third Example

1) Case one: In figure 4, The two algorithms gave good contours from the image in the left. For the encoder decoder output, the algorithm gave us the contours of the girl in the image. For the PbLite algorithm output, The output show the contours of the girl in the image in addition to the contours of the stairs.

2) Case Two: In figure 5, We can see a boy sitting beside the tool in the image. For the output of the encoder decoder network, the network predicted good contours representing the boy and his tool. On the other hand for the PbLite, the algorithm failed to detect the contours of the boy, so instead the algorithm detect alot of contours in the image and failed to generalize these contours into a bigger one.

3) Case Three: In figure 6, We can see a mountain and some trees down this mountain and the sky. For the encoder decoder network, it failed to detect any object contour and the output is almost empty. On the other hand, The Pblite algorithm succeeded to detect the contours of the mountain and the trees in the image.

Image Segmentation

For both algorithms, we ran the algorithm over the 200 hundred test images in the BSDS500 dataset. Each algorithm has a different output style than the other algorithm. For the meanshift algorithm, The output image contains the same color the original image have. For the FCN-ResNet101 output, the output contains diffornt color for each object.

Fig. 7. Meanshift output

4) Segmentation using the meanshift algorithm: In figure 7, The algorithm tried to divide the image into different clusters using the color and location of each pixel.

Fig. 8. FCN-ResNet101 output

5) Segmentation using the FCN-ResNet101: The algorithm is used for semantic segmentation. The network returns the the background as black and for each object category in the image a different color. For example in figure 8, both horses are colored with same color and the background is black. Another observation is that the network returns a reshaped image, this problem can be solved by reshaping the output image using the shape of the original image.

V. Conclusion

Contour Detection

For contour detection part, we implemented two different algorithms. One approach was to use non-machine learning algorithm. The second approach is to make use of deep learning techniques.

For the first approach, the PbLite algorithm which overcomes the problems that Sobel and Canny algorithms have. The algorithm tries to extract the difference in the image texture, the brightness and the color, then combine the results of these steps and combine the results with the Sobel and Canny output to extract the contours of the image.

For the second approach, the encoder decoder network which tries to extract the features of the image using the VGG16 network, and then decode the feature into contour image using the decoder network mentioned in III-A. The VGG16 network contains 16 layer with total 138 million parameters. The first layer of decoder “de-conv6” is designed for dimension reduction that projects 4096 “de-conv6” to 512-d with 1×1 kernel so that the pooling switches from “conv5” can be re-used to upscale the feature maps by twice in the “deconv5” layer. The number of channels of every decoder layer is properly designed to allow unpooling from its corresponding maxpooling layer. All the decoder convolution layers except “deconv6” use 5×5 kernels. All the decoder convolution layers are followed by relu activation function except the one next to the output label. The network was trained using the PASCAL VOC dataset with refined hand drawn contour images.

After comparing the two output for each algorithm, we can conclude that neither non machine algorithms nor the CNN algorithms can solely solve the problem of contour detection. For the PbLite algorithm, the algorithm failed to detect contours in the images having too much details in it. While for the encoder decoder network, the network failed to predict the contours in the images having no dominant object in the image.

For Future work, we can fuse both algorithms into one big system. We can make use of the details the PbLite algorithm, and use the encoder decoder network to detect the most probable contours and then introduce new system that we can rely on in detecting the contours.

Image Segmentation

Same as the contour detection, we tried two different algorithms for image segmentation, meanshift segmentation, and FCN-ResNet101 pre-trained model. Both algorithm deliver a satisfying output but with different style.

To perform the meanshift clustering algorithm, we used the famous machine learning library (sklearn). each pixel of the images were represented as independent entry, the pixel were represented using its row, column, red channel, green channel and the blue channel. The values of each attribute were normalized to get better performance. After normalization, sklearn library were used to estimate the bandwidth values. The normalized data were used in the meanshift clustering implemented by sklearn. After clustering the pixels, the image were reshaped to their original shape.

FCN-ResNet-101 is a high-quality model for semantic segmentation. FCN-ResNet-101 uses the ResNet101 as an encoder. We used a pre-trained model implemented in the pytorch framework. The pre-trained model was trained on the COCO dataset which contains 20 categories that are present in the Pascal VOC dataset.

After comparing the results of each algorithm, we observed that:
For the meanshift, The algorithm tried to divide the image into different clusters using the color and location of each pixel. For FCN-ResNet101, The network returns the the background as black and for each object category in the image a different color

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